

A Fundamental-Analysis-Based Test for Speculative Prices

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ABSTRACT: I investigate the possibility that recent price movements include significantly larger speculative components than those observed historically, where speculation is defined as the component of price that does not co-move with fundamentals. Specifically, at the aggregate level, price and accounting fundamentals co-move historically (1979–1993) but do not co-move recently (1994–2008). The lack of co-movement in recent periods is accompanied by a significant reduction in the positive association between ratios of accounting fundamentals-to-price with future market returns. Changes in measurement error in accounting fundamentals do not appear to cause the lack of co-movement in recent periods, and risk- and growth-based explanations are not supported by the data. The results of this study provide evidence of a structural change in the long-run association between price and accounting fundamentals.

Keywords: *fundamental analysis; speculative markets; co-movement; forecasting.*

Data Available: *All data are available from the sources described in the text.*

I. INTRODUCTION

This study investigates the possibility that recent price movements include significantly larger speculative components than observed historically, defining speculation as the component of price that does not co-move with fundamental value ([Harrison and Kreps](#)

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1978). Recent periods labeled “bubbles,” “crashes,” and “financial crises” have re-ignited academic and policy debates about whether price can deviate from fundamental value for an extended period of time. This question is of interest to policymakers, as research in this area could suggest potential directions for pre-emptive actions aimed at deflating potential asset price bubbles (Hunter et al. 2005). Central to this debate is the question of whether large shifts in prices are due to either shifts in fundamentals or departures of prices from fundamentals (Brunnermeier 2001, 47). Measuring the time-series properties of price- and accounting-based measures of fundamental value is a crucial first step in understanding this important policy question. The results of this study contribute to this debate by documenting evidence of a structural change in the time-series association between prices and accounting fundamentals in recent time periods.

My study extends the work of Lee et al. (1999), who provide evidence of co-movement between price and accounting fundamentals, using estimates from residual income models, over a period (1979–1996) that precedes the recent periods labeled “bubbles,” “crashes,” and “financial crises.” I examine whether their results continue to hold in recent periods. Following Lee et al. (1999), I construct monthly aggregate indices of price and accounting fundamentals and test for co-movement at the aggregate level.¹ In contrast to Lee et al. (1999) I do not find evidence of co-movement between price and fundamental value over the period 1979–2008; evidence of co-movement, however, is found in the earlier period ending in 1996 which overlaps with Lee et al. (1999). These results provide evidence that the way in which the market prices accounting fundamentals has structurally changed in more recent years.

I also document evidence of a structural change in the association between ratios of accounting fundamentals-to-price with future market returns documented by Lee et al. (1999). Specifically, I find no evidence of a positive association between ratios of accounting fundamentals-to-price and future market returns in the 1994–2008 period, while in the 1979–1993 period the association is consistent with that in Lee et al. (1999). This result presents evidence of a structural change in the time-series properties of price. Using a multivariate time-series approach, I find evidence that the lack of co-movement appears to be driven primarily by changes in the short-term time-series properties of price.

These results should be interpreted with the important caveat that financial statement analysis provides numerous methods for estimating the fundamental value of a stock based on accounting inputs (Penman 2007). As my study focuses on identifying possible changes in the time-series relation between prices and accounting fundamentals in recent time periods, I use a residual income model that combines historical earnings and book-value information in my primary analysis. The residual income model can also accommodate forward-looking information from analysts’ forecasts, as well as changes in expected risk and expected growth assumptions, which I investigate in a supplemental robustness analysis.

In the robustness analysis, I document that risk- and growth-based explanations for the lack of co-movement are not supported by the data, and that including analysts’ forecasts in the residual income model does not explain the lack of co-movement in recent periods. I also provide evidence of the timing of the structural break in the co-movement between prices and accounting fundamentals using a rolling-windows estimation technique. Using this approach, I first document a lack of co-movement between prices and fundamentals in 1996. These results suggest that periods commonly labeled “bubbles,” “crashes,” and “financial crises” can potentially be identified as

¹ Co-movement can be formally tested using cointegration analysis, which is the statistical term that describes the long-run equilibrium tendency of two variables to be tied together (Hamilton 1994). When examining nonstationary (i.e., not mean-reverting) variables, cointegration is an appropriate method; variance ratios, which have been used in prior research, are not appropriate as variance is undefined for nonstationary variables (Kleidon 1986).

periods for which prices and accounting fundamentals do not have a tendency to co-move at the aggregate level.²

My results have implications for the literature on market-based research that investigates the association between market prices (or returns) and accounting information at the aggregate level. Specifically, OLS-based estimates of the relation between aggregate time-series of price and accounting variables can include spurious components. Future research conducted at the aggregate level could investigate which factors influence the tendency of prices and accounting fundamentals to co-move, furthering the understanding of their lack of co-movement in recent periods. Future researchers could also investigate whether there is a lack of co-movement, or variation in the cross-section at the firm level, as a lack of co-movement at the aggregate level does not automatically imply a lack of co-movement at the firm level (Gonzalo 1993). Finally, my results suggest that future research analyzing market prices and returns with accounting fundamentals should investigate the role of trends in the data to avoid incorrect inferences.

Section II develops my model and hypothesis. Section III presents the main tests, Section IV provides additional forecasting and multivariate analysis, and Section V presents robustness checks. Section VI concludes.

II. MODEL AND HYPOTHESIS

The Expected Long-Run Relation between Price and Measures of Fundamental Value

The tests I perform in this study are based on the market fundamentals hypothesis, which states that price reflects the present value of future dividends.³ Errors in prices arising from transaction costs, trading frictions, and mispricing prevent market efficiency from holding exactly at every possible date (Garman and Ohlson 1981; Lee et al. 1999). A generalization of the fundamentals hypothesis, allowing for frictions and temporary mispricing, states that prices should co-move with their fundamental values (Campbell and Shiller 1988; Lee et al. 1999; Ritter and Warr 2002). The formal test for co-movement, cointegration analysis, is appropriate in this case, as (1) accounting metrics do not incorporate all value-relevant information immediately (Collins et al. 1994; Weiss et al. 2008), (2) if the market misprices accounting information, then the correction of this mispricing is not expected to be immediate due to frictions or limits to arbitrage (Ou and Penman 1989; Sloan 1996; Maines and Hand 1996; Shleifer and Vishny 1997; Rangan and Sloan 1998), and (3) there are strong theoretical links between current accounting measures and the present value of future cash-flows (Ohlson 1995; Penman 2007).

To formulate the expected association between price and fundamental values, I begin with the assumption that the “true” fundamental value, typically referred to as intrinsic value, is unobservable and equal to the present value of all future dividends:

² As my sample is aggregated, my results are not heavily affected by the increase in the number of smaller firms over time, which suggests that even large established firms’ prices can deviate from their fundamentals for an extended period. In contrast, much of the debate regarding the existence of significant speculation in market prices has been limited to the examination of Internet stocks that were listed on the NASDAQ during the late 1990s. For example, using data from the late 1990s, Ofek and Richardson (2002, 2003) and Demers and Lev (2001) highlight the “implausibly high” growth rates needed to justify the prices of Internet stocks. Bartov et al. (2002) show an association between Internet stock prices and nontraditional measures of potential value, and Cooper et al. (2001) show that firms that changed their names to include “.com” earned abnormal market returns. Finally, Pastor and Veronesi (2006) argue that the “bubble- and crash-like” appearance of stock prices was due to the resolution of previously uncertain growth expectations for Internet firms.

³ See, for example, LeRoy and Porter (1981), Blanchard and Watson (1982), Summers (1986), and Shiller (1989).

$$V_t^* = \sum_{\tau=1}^{\infty} E_t(R_t^{-\tau} D_{t+\tau}), \quad (1)$$

where V_t^* is the unobservable intrinsic value of the stock assumed to be a random walk (Samuelson 1965), and $R_t^{-\tau}$ is 1 plus the discount rate, which is time-varying and assumed to be greater than zero (Campbell and Shiller 1988). $D_{t+\tau}$ equals the real dividend paid to the owner of the stock between $t-1$ and t , and E_t denotes expectations conditional on information available at time t .

Following Lee et al. (1999), I consider price to be an estimate of intrinsic value with error. In log form, the expected relation between log price (p_t) and (unobservable) log intrinsic value (v_t^*) is:

$$p_t = \gamma_1 v_t^* + u_{1t}^*, \quad (2)$$

where $p_t = \log(P_t)$, $v_t^* = \log(V_t^*)$, and u_{1t}^* is unobservable mispricing error.

Fundamental analysis uses accounting fundamentals like earnings and book-values to estimate fundamental value, with numerous estimation techniques available, such as the residual income model (Ohlson 1990, 1995). The measurement of fundamental value using accounting inputs or forecasts of accounting inputs provides a second measure of intrinsic value with error:

$$f_t = \gamma_2 v_t^* + u_{2t}^*, \quad (3)$$

where $f_t = \log(F_t)$, F_t is an estimate of fundamental value using accounting fundamentals to approximate the unobservable intrinsic value in Equation (1), and u_{2t}^* is unobservable measurement error.

In Equations (2) and (3), both price and accounting fundamentals are measures of intrinsic value with error, and they share the unobservable intrinsic value as a common trend. The time-series relation between accounting fundamentals and price can be written by substituting the unobservable intrinsic value for price in Equation (3), yielding:

$$f_t = \beta p_t + \varepsilon_t, \quad (4)$$

where $\beta = \gamma_2 \gamma_1^{-1}$ and $\varepsilon_t \equiv u_{2t}^* - \gamma_2 \gamma_1^{-1} u_{1t}^*$. Equation (4) states a general representation of the market fundamentals hypothesis and has the following unique property: if ε_t is a stationary process, then β characterizes the long-run or “equilibrium” relation between price and fundamental value.

If β does characterize the equilibrium relation between price and fundamental value, then if $f_t < \beta p_t$, then the expectation is that, over time, either accounting fundamentals will increase or price will decrease to restore equilibrium; the converse is also true if $f_t > \beta p_t$. This process of either price or accounting fundamentals correcting toward the equilibrium relation implies that the error term in Equation (4), ε_t , is expected to mean-revert over time, which can be written as:

$$\varepsilon_t = \rho \varepsilon_{t-1} + e_t, \text{ for } |\rho| < 1, \quad (5)$$

where e_t is a mean zero random variable. As the time-series mean of ε_t does not theoretically equal zero, the times-series could potentially include a permanent difference between the levels of price and accounting fundamentals. Including a constant, a_0 , in the model allows for tests of stationarity with a permanent difference between the two variables (i.e., ε_t oscillates around a constant level). It is also possible that there is a stable increasing difference between price and accounting fundamentals, and including a time-trend, δt , in the model allows for tests of stationarity around an increasing difference over time (i.e., ε_t oscillates around a constant time-trend). For example, unrecognized intangible assets could lead to both permanent and increasing differences between prices and accounting fundamentals. Adding a constant and a time trend to the model yields:

$$\varepsilon_t = a_0 + \delta t + \rho \varepsilon_{t-1} + e_t, \quad (6)$$

where a_0 is a non-zero constant and δt is the time-trend.

The Expected Properties of the Log Fundamental-Value-to-Price Ratio

Lee et al. (1999) provide evidence that ratios of accounting fundamentals-to-price are stationary around an intercept and stationary around an intercept and time-trend, using the residual income model. Evidence of stationarity in fundamental-to-price ratios indicates that accounting fundamentals and price are cointegrated. Equations (4) and (6) can be reconciled to Lee et al. (1999) by setting $\beta = 1$, as the assumption that $\beta = 1$ allows Equation (4) to be represented as the log fundamental-to-price ratio; i.e., $f_t/p_t = \varepsilon_t$.⁴ As the measurement errors in price and accounting fundamentals may require more than one month to correct, the errors from Equation (6) are likely to be serially correlated. The model is therefore estimated in changes. Thus, the following regression models, which include a constant and time-trend, test whether the times-series is stationary (see Appendix A for further details):

$$\Delta \varepsilon_t = a_0 + (\rho - 1)\varepsilon_{t-1} + e_t, \text{ and} \quad (7)$$

$$\Delta \varepsilon_t = a_0 + \delta t + (\rho - 1)\varepsilon_{t-1} + e_t, \quad (8)$$

where Equation (7) includes a constant and Equation (8) includes both a constant and a time-trend. The null in both regressions is that the variable ε_t has a unit root (i.e., it is nonstationary) when $\rho = 1$. If $\rho \geq 1$, then there is evidence that accounting fundamentals and price are not cointegrated.⁵ Alternatively, $\rho < 1$ would indicate evidence of cointegration between accounting fundamentals and price, as the changes in errors, on average, are reducing the prior level of the disparity (as $(\rho - 1 < 0)$ in this case). Evidence of cointegration is important as it implies that deviations between prices and fundamentals decay relatively quickly, suggesting prices and accounting fundamentals both reflect intrinsic value in a timely manner. I report two formal test statistics: an adjusted regression coefficient labeled Z-Rho, which is calculated by adjusting the estimate of $T^*(\rho - 1)$, where T is number of time-series observations used in the estimation, and Z-Tau, which is an adjusted t-statistic related to the $(\rho - 1)$ coefficient estimate.⁶ If $\rho < 1$, then both Z-Rho and Z-Tau will be negative, and, in general, are more negative for lower estimates of ρ where lower values of ρ imply less persistent, or faster decaying, errors. These statistics follow the distribution provided in Phillips and Perron (1988).

Stability of the Relation between Price and Value

The primary objective of this study is to investigate whether price and accounting fundamentals remain co-integrated in recent periods labeled “bubbles,” “crashes,” and “financial crises.” A lack of co-integration suggests that speculation is not removed from prices in a timely manner, where speculation is defined as a component of price that does not co-move with fundamental value (Harrison and Kreps 1978). Speculation cannot be directly observed; however, an increase in the time that speculation takes to decay in recent periods has implications for the tendency of accounting

⁴ As both price and accounting fundamentals are in logs, the ratio of logs is equivalent to differences in logs.

⁵ If $\rho > 1$, then the time-series is nonstationary and is considered “explosive” in nature.

⁶ The adjustments to these statistics are functions of T , the determinants of the cross-product matrix and the difference in the squared sum less the squared errors of the model (Maddala and Kim 2002, 80–81). As the statistics depend on T , it is possible that more frequently observed fundamentals data, or a longer time span of data sampled less frequently may increase the power of finding cointegration. I examine the first explanation in my robustness analysis and leave the second possibility for future research.

fundamentals and price to co-move. Specifically, co-movement relies on the speed at which measurement errors in accounting fundamentals and price decay.⁷ A testable prediction is that there is a change in recent periods in the stationarity of ratios of accounting fundamentals-to-prices:

Hypothesis: The time-series dynamics of ratios of accounting fundamentals-to-price change from being stationary historically to being nonstationary in recent periods.

To test my hypothesis, I begin by defining a “recent” period and a “historical” period, which I split into the first half (1979–1993) and second half (1994–2008) of the available times-series to increase the power of the test.⁸ The first period ends prior to the IPO of Netscape, and the second period includes the “bubble” of the late 1990s, the “crash” of 2001, and recent “financial crisis” periods. To test the hypothesis, I estimate Equations (7) and (8) across the two time periods. For example, for Equation (8):

$$\Delta \varepsilon_t = \begin{cases} a_0 + \delta t_0 + (\rho_0 - 1)\varepsilon_{t-1} + e_t, & t = 1, \dots, T, \\ a_1 + \delta t_1 + (\rho_1 - 1)\varepsilon_{t-1} + e_t, & t = T + 1, \dots, T + m, \end{cases} \quad (9)$$

where ρ_0 is estimated from the first T observations and ρ_1 is estimated from the last m observations. The log fundamental-to-price ratio, ε_t , is stationary when the estimate of $\rho < 1$; conversely, the log fundamental-to-price ratio is nonstationary when the estimate of $\rho \geq 1$. There are four possible outcomes when comparing historical and recent periods. Specifically, the fundamental-to-price ratio (1) is nonstationary in both periods ($\rho_0 \geq 1, \rho_1 \geq 1$), (2) is stationary in both periods ($\rho_0 < 1, \rho_1 < 1$), (3) is nonstationary in the historical period and stationary in the recent period ($\rho_0 \geq 1, \rho_1 < 1$), or (4) is stationary in the historical period and nonstationary in the recent period ($\rho_0 < 1, \rho_1 \geq 1$). Only case (4) provides evidence in support of the hypothesis.

III. EMPIRICAL ANALYSIS

Data Sources and Sample Selection

I include all firms in the sample covered by both CRSP and Compustat over the period 1979–2008; I do not require that firms have available data for the entire period, and I exclude observations for which the book-value of equity is negative.⁹ Observations where residual income is negative (where earnings are less than the discount rate times book-value) are retained, but with the residual income model estimate set equal to book-value.¹⁰ I collect financial data required for measures of fundamentals from the Compustat database (annual book-values, earnings, and dividends), and

⁷ Both larger and smaller levels of speculation may cause a lack of co-movement if they tend not to decay quickly during the sample period.

⁸ Lee et al. (1999) provide evidence of cointegration for a time-series ending in June 1996. While a split in the middle of my time-series may appear arbitrary, testing for cointegration using short time spans arguably biases toward finding no cointegration due to the decreased power of the test. Podivinsky (1998) suggests that the critical values for cointegration tests may be inappropriate for sample sizes of less than 100 observations. I split the sample equally, from 1979 to 1993 and from 1994 to 2008, yielding 180 observations in each sample period, so that the power of the test is not a concern in these comparisons. I consider alternative splits of the time-series in my robustness analysis.

⁹ I obtain similar results when limiting the sample to firms that have available data in the Dow 30, the S&P 500, the top 1000 or top 50 by market capitalization. I investigate the restriction of the sample to firms with analyst coverage in my robustness analyses.

¹⁰ Due to the increasing frequency of loss firms over the estimation period, I do not exclude these firms from the analysis. The choice to replace negative residual income with book-value rules out the estimation of implausible negative value estimates. This approach also implicitly assumes that losses are transitory and that the firm is valued at its adaption value (Burgstahler and Dichev 1997). Results are not sensitive to the exclusion of loss firms.

monthly market-value of equity data using month-end share price and shares outstanding on CRSP. I source interest rate data from the Treasury constant-yield-to-maturity bond series. Following [Lee et al. \(1999\)](#) I use the same publicly available financial information over multiple months in the construction of monthly accounting fundamental estimates. For example, I retain book-value in monthly observations from four months after the financial year-end, until three months after the subsequent financial year-end when it is updated.¹¹ I aggregate all firm-level observations of accounting fundamentals and market values at the end of the month to construct the monthly times-series.¹²

Measurement of Fundamental Value

I measure accounting fundamentals as historical dividends, d (Compustat DVC), and historical book-values, b (Compustat CEQ), and complement them with valuations estimated from the residual income model. Firms that do not pay dividends are retained in the aggregate dividend measure and corresponding aggregate price measure. Following prior research, I implement the residual income model using two main approaches, the forecast approach and the linear information dynamic approach. The forecast approach uses the structure of the residual income model to incorporate explicit estimates of future dividends (e.g., [Frankel and Lee 1998](#); [Lee et al. 1999](#)). Using the clean surplus relation, the forecast-based residual income model (vf) can be implemented using the following structural form with T -period-ahead observations of earnings forecasts:

$$vf(T)_t = b_t + \frac{f(1)_t - r \cdot b_t}{(1+r)} + \frac{f(2)_t - r \cdot b(1)_t}{(1+r)^2} + \dots + \frac{f(T)_t - r \cdot b(T-1)_t}{(1+r)^{T-1}r}, \quad (10)$$

where b_t is the book-value of equity, and $f(\cdot)$ is the forecast of earnings for periods $\{1, 2, \dots, T\}$, r is the equity cost of capital. The model can then be collapsed to provide a one-period estimate of fundamental value similar to the Gordon growth model by using a single forecast of earnings and assuming a perpetual growth rate g . Specifically:

$$vf(x)_t = b_t + \frac{f(1)_t - r \cdot b_t}{r - g}. \quad (11)$$

This very simple structure can be used to calibrate multiple measures of value, based on changing $f(1)$, r , and g .¹³ In implementing this model, I assume that r equals the three-year Treasury constant-yield-to-maturity treasury bond rate plus an equity premium of 6 percent, and that g is set equal to 3 percent. Book-value, b_t , is the end of year book-value from the most recent fiscal year-end (Compustat CEQ), and $f(1)$ is the one-year forecast of earnings. To proxy for $f(1)$, I use the most recent historically available annual earnings as the forecast of earnings (Compustat IB), which assumes a random walk in annual earnings.

The second approach is to consider the linear information dynamics suggested by [Ohlson \(1995\)](#) to combine the information in earnings and book-value ([Dechow et al. 1999](#); [Myers 1999](#)):

$$vl(x)_t = b_t + \alpha_1 x_t^a + \alpha_2 v_t, \quad (12)$$

¹¹ In untabulated results, the main results are not sensitive to the use of quarterly book-values, dividends, and earnings.

¹² In untabulated results, I find that the main results are consistent when measuring the time-series using either quarterly or annual frequencies rather than monthly. The conclusion drawn from the relevant econometrics literature is that for a given time span, it is preferable to use higher frequency data when it is available ([Maddala and Kim 2002](#), 229). Thus, I follow [Lee et al. \(1999\)](#) and focus on monthly frequencies.

¹³ I consider the impact of changes to assumptions regarding r , g , and $f(1)$ in a subsequent robustness analysis.

where $\alpha_1 = \frac{\omega}{(1+r-\omega)} \cdot \frac{\alpha_2=(1+r)}{(1+r-\omega)(1+r-\gamma)}$, $x_t^a = ni_t - r \cdot b_{t-1}$, where ni_t is annual net income ending at time t , and v_t is other information (Ohlson 1995). Equation (12) is based on the following assumed linear information dynamics:

$$x_{t+1}^a = \omega x_t^a + v_t + e_{1,t+1}, \quad (13)$$

$$v_{t+1} = \gamma v_t + e_{2,t+1}. \quad (14)$$

When other information (v_t) is assumed to equal 0, and $0 < \omega < 1$, the model comprises a convex combination of book-value and residual income:

$$vI(x)_t = b_t + \frac{\omega_t x_t^a}{(1 + r_t - \omega_t)}, \quad (15)$$

where ω and γ are constants. Following Dechow et al. (1999), I estimate these terms using a one-period lagged AR(1) model, using the prior year's data for each industry.

Descriptive Statistics

Table 1 presents summary statistics for the sample period. Panel A presents descriptive statistics for the market portfolio price and fundamentals. The log of price, p , is on average 15.76, with a standard deviation of 0.75; the log of the one-period-ahead perpetuity of residual income, $vf(x)$, has a mean of 15.35 and a standard deviation of 0.69. I also provide stationarity tests for all of the variables in Panel A, finding that all variables are nonstationary in levels and stationary in first differences (i.e., $I(1)$ processes).

Table 1, Panel B presents descriptive statistics of the log fundamental-to-price ratios. The log ratio of $vf(x)/p$ has a mean of -0.41 , suggesting that aggregate value is, on average, 66.4 percent of aggregate price. Panel B documents that all of the first-order autocorrelations for the fundamental-to-price ratios are high, suggesting they are either nonstationary or slowly mean-reverting.¹⁴ The first-order autocorrelation coefficient reported in Panel B for $vf(x)/p$ is 0.948, which is higher than the 0.93 reported by Lee et al. (1999) for a similar model. Panel C reports descriptive statistics of the log fundamental-to-price ratios split into two equal subperiods, 1979–1993 and 1994–2008. There is some evidence that is consistent with larger deviations of price from accounting fundamentals in the 1994–2008 period, with many of the standard deviations and first-order autocorrelations of the ratios increasing. This evidence, however, is inconclusive because if the ratios are nonstationary, then the standard deviations are not interpretable (Kleidon 1986). Since the first-order autocorrelation is close to 1, and there is possible evidence of differences in the fundamental-to-price ratios across the two periods, formal statistical tests are required to determine if $vf(x)/p$ is stationary in recent periods.

Figure 1 presents the time-series variation of ratios of aggregate price-to-fundamentals for book-value and $Vf(x)$. From 1979 to the end of the Lee et al. (1999) time period (June 1996), the fundamental-to-price ratios fluctuate regularly within a tight band, suggesting mean-reversion. As in Lee et al. (1999), the crash of 1987 is apparent in the ratios. Following the Lee et al. (1999) time period, I find that all ratios are much more volatile and appear to support the observations of a “bubble,” a “crash,” and a “financial crisis.” Specifically, the extent to which the ratios increase from June 1996 until March 2000 is unprecedented in this times-series, as is the steepness of the declines in the price-to-fundamentals ratios beginning in April 2001. At the end of the times-series in late 2008,

¹⁴ To some extent the high autocorrelations of the book-to-price and dividend-to-price ratios could be driven by the retention of annual accounting variables for 12 periods.

TABLE 1
Summary Statistics

Panel A: Descriptive Statistics for the Aggregate Market Portfolio ($T = 360$)

Variable	Mean	Std. Dev.	Min.	Max.	Stationarity Tests of Logged Variables			
					In Levels		In Changes	
					Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)
p	15.76	0.75	14.45	16.82	-1.8	-1.1	-317.4*	-17.1*
b	14.96	0.44	14.37	15.83	0.1	0.1	-347.0*	-18.4*
d	12.02	0.37	11.44	12.77	0.9	0.8	-365.5*	-20.3*
$vf(x)$	15.35	0.69	14.39	16.89	1.5	1.0	-296.6*	-15.6*
$vl(x)$	15.01	0.47	14.37	15.91	1.2	1.2	-366.5*	-20.3*

Panel B: Descriptive Statistics of Fundamental-to-Price Ratios ($T = 360$)

Variable	Mean	Std. Dev.	Min.	Max.	Autocorrelation at Lag					
					1	12	24	36	48	60
d/p	-3.74	0.43	-4.55	-2.85	0.989	0.888	0.805	0.694	0.564	0.467
b/p	-0.79	0.37	-1.54	0.00	0.983	0.851	0.719	0.601	0.466	0.394
$vf(x)/p$	-0.41	0.26	-1.10	0.68	0.948	0.592	0.384	0.225	0.118	0.091
$vl(x)/p$	-0.75	0.35	-1.48	0.00	0.949	0.566	0.363	0.165	0.044	0.001

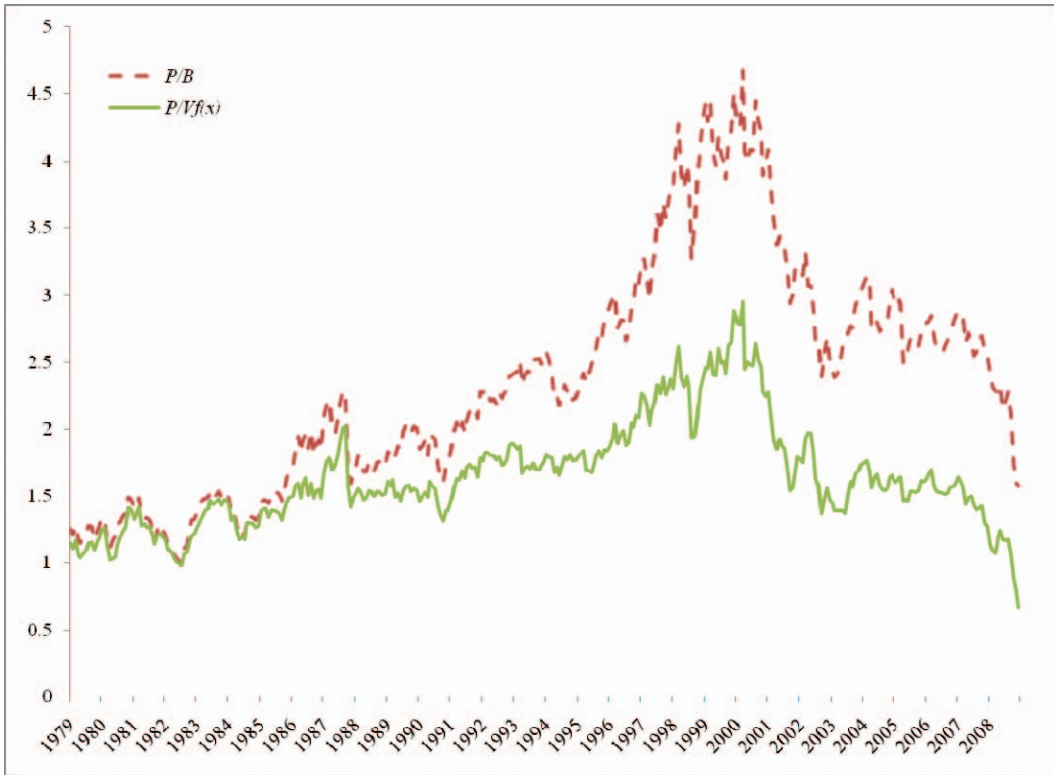
Panel C: Descriptive Statistics for Fundamental-to-Price Ratios by Period

Variable	Mean	Std. Dev.	Min.	Max.	Autocorrelation at Lag					
					1	12	24	36	48	60
(1) Historical period (January 1979–December 1993, $T = 180$)										
d/p	−3.37	0.23	−3.80	−2.85	0.944	0.536	0.287	−0.059	−0.260	−0.375
b/p	−0.50	0.24	−0.93	0.00	0.928	0.607	0.316	0.084	−0.131	−0.246
$vf(x)/p$	−0.33	0.14	−0.68	0.02	0.938	0.534	0.260	0.077	−0.002	−0.007
$vl(x)/p$	−0.48	0.23	−0.87	0.00	0.949	0.572	0.316	0.092	−0.016	−0.059
(2) Recent period (January 1994–December 2008, $T = 180$)										
d/p	−4.10	0.23	−4.55	−3.50	0.964	0.619	0.450	0.285	0.130	0.031
b/p	−1.09	0.21	−1.54	−0.46	0.972	0.685	0.433	0.263	0.122	0.072
$vf(x)/p$	−0.50	0.31	−1.10	0.68	0.934	0.466	0.421	0.313	0.143	0.001
$vl(x)/p$	−1.01	0.21	−1.48	−0.34	0.923	0.380	0.412	0.290	0.134	−0.020

* $p < 0.05$.

Panel A reports the log of the aggregate price (p), dividends (d), book-values (b), and logged accounting-based value estimates where $vf(x)$ is the one-period forecast-based residual income model and $vl(x)$ is the residual income model based on the linear information dynamics of [Ohlson \(1995\)](#). Panels B and C report ratios of the aggregate value estimate divided by the aggregate price in logs. The stationarity estimates are performed using the Phillips-Perron estimates with a constant and one lag, Equation (7) in the text. All variables are aggregated on a monthly basis using available data. T is the number of time-series observations.

FIGURE 1
The Time-Series of the Ratio of Price-to-Value

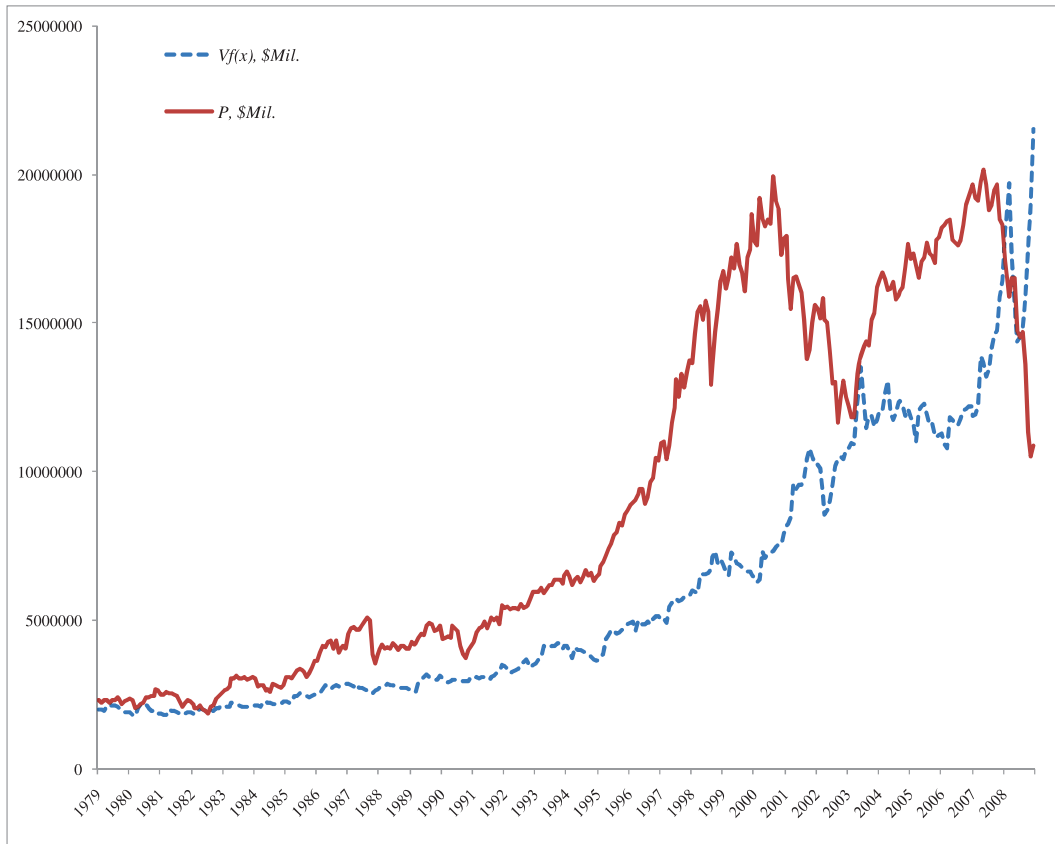


This graph displays ratios of price-to-book-value and estimates of the residual income model. Aggregate (P) is divided by either (1) aggregate book-value (B) or (2) aggregate accounting fundamentals using $Vf(x)$, which is measured using the sum of book-value and a perpetuity of one-period-ahead residual income using historical earnings growing at 3 percent, discounted using the three-year Treasury constant-yield-to-maturity bond rate plus a 6 percent equity premium, and book-value is measured as total common equity. These ratios are constructed as aggregate indices and represent the sum of all firms with sufficient data during the period 1979 to 2008.

the financial crisis is evident in the data, which shows a sharp decline in all of the price-to-fundamentals ratios. The larger oscillations in recent periods provide preliminary evidence consistent with a change in the time-series relation between accounting fundamentals and price. Formal analysis of stationarity for each ratio is important, as it tests whether there is statistical evidence of co-movement between accounting fundamentals and price (e.g., [Engle and Granger 1987](#)).

Figure 2 graphs the times-series of aggregate price, P , and accounting fundamentals using $Vf(x)$ separately, to provide additional insight into possible changes in the properties of the price and accounting fundamentals in recent periods. Prior to 1995, both times-series drift upward at approximately the same rate. The “bubble period” is clearly evident in the data where P diverges from $Vf(x)$ in the late 1990s. The “crash” of 2001 precedes a short period in the mid-2000s where P and $Vf(x)$ appear to be closely aligned, similar to the historical relation. The “financial crisis” is evident in the substantial declines in P at the end of the sample period, and the large effect of low

FIGURE 2
The Time-Series of Price and Value



This graph displays the time-series of aggregate market value (P) and aggregate accounting fundamentals using $Vf(x)$, which is measured using the sum of book-value and a perpetuity of one-period-ahead residual income using historical earnings growing at 3 percent, discounted using the three-year Treasury constant-yield-to-maturity bond rate plus a 6 percent equity premium. Each index is constructed as an aggregate index reported in millions of U.S. dollars and representing the sum of all firms with sufficient data during the period 1979 to 2008.

interest rates on the $Vf(x)$ model.¹⁵ Importantly, over the entire period represented in the figure, there is a strong upward trend in both the P and $Vf(x)$ measures. Because there is evidence of similar upward trends in nonstationary variables, multivariate cointegration analysis is especially important, as it provides a test of whether a relation between two variables exists due to a common trend, which in this case is whether prices and accounting fundamentals share the trend of intrinsic value (e.g., [Stock and Watson 1988](#)).

¹⁵ $Vf(x)$ appears to be substantially above P in December 2008, but it is the relation between the time-series changes of the individual variables that are important, not their relative levels, which can be modified by changing the equity premium. For example, in results not tabulated here, price and value appear to be approximately equal to 1 for the constant 12 percent discount rate model in December 2008.

Testing for Stationarity in the Log Value-to-Price Ratio

Following Lee et al. (1999), I formally test for stationarity in the log fundamental-to-price ratios using Phillips-Perron tests.¹⁶ Table 2 reports the results of the Phillips-Perron tests of stationarity for accounting fundamental-to-price ratios. Panel A presents the tests with an intercept but without a time-trend (Equation 7); in Panel B the tests include a time-trend (Equation 8). In Columns (1) and (2) of Panel A, the Z-Rho and Z-Tau statistics suggest that all of the fundamental-to-price ratios are nonstationary in the 1979–2008 period.¹⁷

To formally test my hypothesis, I provide Phillips-Perron estimates of stationarity of the accounting fundamentals-to-price ratios for two equal subperiods, 1979–1993 and 1994–2008. In Columns (3) and (4) of Panel A in Table 2, I report the test statistics for the earlier 1979–1993 subperiod. I do not find evidence of stationarity for the b/p and d/p ratios, which is similar to the findings of Lee et al. (1999), or for the ratio that uses the linear information dynamics model to price, $vl(x)/p$, which is similar to the findings in Morel (2003). It is possible that these ratios can be considered stationary in this time period around a trend, which I investigate in Panel B.

Alternatively, there is evidence of stationarity for the $vf(x)/p$ model during the earlier time period (1979–1993).¹⁸ Specifically, in Columns (3) and (4), the $vf(x)/p$ ratio is stationary at the 5 percent level ($Z_\rho = -19.94$, $Z_\tau = -3.24$). In untabulated results, the estimate of the autoregression parameter (i.e., ρ in Equation (7)) is 0.958. This implies that the $vf(x)/p$ ratio has a half-life of 16.1 months, suggesting that deviations from the mean $vf(x)/p$ take over a year to decay.¹⁹ In direct contrast, when examining the more recent period 1994–2008, I find that the $vf(x)/p$ fundamental-to-price ratio is nonstationary based on the statistics reported in Columns (5) and (6), where $Z_\rho = 0.89$ and $Z_\tau = 0.31$. In untabulated results, the autoregression parameter 0.994 implies that the $vf(x)/p$ ratio has a half-life of 115.2 months, approximately a 700 percent increase from the prior period. Using a Chow test, I test for a structural break using December 1993 as the breakpoint, to assess statistical differences between the two periods. I present a summary indicator of evidence of structural change between the two time periods at the 5 percent level in Column (5) with a dagger symbol. I find evidence of structural change for $vf(x)/p$, which confirms a significant change across the two time periods in the tendency of this fundamental-to-price ratio to mean-revert.

One possible reason for finding a lack of co-movement for the book-value, dividend, and $vl(x)$ based ratios is the omission of the trend term. Panel B reports stationarity tests that include a time-trend term in the autoregression model (Equation 8). Given the upward trend in price observed in the earlier part of the times-series (see Figure 2), it is not surprising that the model appears to fit the data in this earlier period. The statistics reported in Panel B provide evidence of cointegration for all of the accounting fundamentals in the earlier time period, reported in Columns (3) and (4), as all of the accounting fundamental-to-price ratios are stationary at the 10 percent level or better. For the more recent 1994–2008 period, the evidence in Columns (5) and (6) suggests that all of the ratios are nonstationary and are structurally different from the earlier period under a Chow test. Interpretation of these tests, however, requires caution, as including a time-trend allows for the

¹⁶ I use the Phillips-Perron test-statistics instead of the augmented Dickey-Fuller test-statistics because the former test corrects for heteroscedastic and serially correlated residuals (Hamilton 1994).

¹⁷ In untabulated results, I obtain similar results when limiting my sample to the Dow 30 firms covered in Lee et al. (1999). My results are similar when I use either equal-weighted or value-weighted (using market value of equity) averages of the firm-specific value-to-price ratios, except that the equal-weighted method does not provide evidence of cointegration around a trend in the 1979–1993 period. Both of these alternative methods display more volatile ratios of price-to-accounting-fundamentals with more pronounced peaks during the late 1990s.

¹⁸ In untabulated results, I find that $vf(x)/p$ ratios for the 1979 to June 1996 period reject the null of nonstationarity at the 5 percent level, confirming that the Lee et al. (1999) results are generalizable to a broader market index.

¹⁹ The half-life is calculated as $\log(0.5)/\log(\rho)$.

TABLE 2
Stationarity Tests of Fundamental-to-Price Ratios

Panel A: Phillips-Perron Tests for Stationarity with a Constant

$$\Delta \varepsilon_t = \alpha_0 + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (7)$$

	1979–2008 <i>T</i> = 360		1979–1993 <i>T</i> = 180		1994–2008 <i>T</i> = 180	
Variable	(1) Z-Rho (Z_ρ)	(2) Z-Tau (Z_τ)	(3) Z-Rho (Z_ρ)	(4) Z-Tau (Z_τ)	(5) Z-Rho (Z_ρ)	(6) Z-Tau (Z_τ)
<i>d/p</i>	−3.78	−1.52	−3.32	−1.08	−3.88	−1.06
<i>b/p</i>	−4.10	−1.58	−3.13	−1.10	−0.90	−0.27
<i>vf(x)/p</i>	−3.36	−0.71	−19.94*	−3.24*	0.89 [†]	0.31
<i>vl(x)/p</i>	−4.30	−1.56	−3.60	−1.21	−0.30	−0.09

Panel B: Phillips-Perron Tests for Stationarity with a Constant and a Time-Trend

$$\Delta \varepsilon_t = \alpha_0 + \delta t + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (8)$$

	1979–2008 <i>T</i> = 360		1979–1993 <i>T</i> = 180		1994–2008 <i>T</i> = 180	
Variable	(1) Z-Rho (Z_ρ)	(2) Z-Tau (Z_τ)	(3) Z-Rho (Z_ρ)	(4) Z-Tau (Z_τ)	(5) Z-Rho (Z_ρ)	(6) Z-Tau (Z_τ)
<i>d/p</i>	−3.82	−0.80	−18.75 ^δ	−3.16 ^δ	−3.44 [†]	−0.96
<i>b/p</i>	−1.84	−0.43	−22.35*	−3.43 ^δ	−2.58 [†]	−0.78
<i>vf(x)/p</i>	−2.70	−0.58	−24.88*	−3.48*	−3.33 [†]	−1.01
<i>vl(x)/p</i>	−1.56	−0.36	−23.47*	−3.52*	−2.47 [†]	−0.73

*, ** $p < 0.05$ and $p < 0.001$, respectively.

^δ $p < 0.1$.

[†] Indicates significant structural change in the model using December 1993 as the breakpoint according to the Chow test (at the $p < 0.05$ level).

This table summarizes the results of the Phillips-Perron unit root tests on the four value-to-price ratios. The unit root tests are performed without a time-trend in Panel A testing Equation (7) in the text and with a time-trend in Panel B testing Equation (8) in the text. Fundamental-to-price ratios are the log of the aggregate accounting fundamentals divided by the log of the aggregate price. Accounting fundamentals are estimated using the following models: *d* is the aggregate annual dividends paid in the most recent fiscal year, *b* is the aggregate book-value measured at the end of the most recent fiscal year, *vf(x)* is the one-period forecast-based residual income model using historical earnings, and *vl(x)* is the residual income model based on the linear information dynamics of [Ohlson \(1995\)](#). The test is against the null of a unit root in the times-series ($\rho = 1$). The test statistics from the Phillips-Perron regressions are an adjusted regression coefficient Z-Rho (Z_ρ) and an adjusted t-statistic Z-Tau (Z_τ).

possibility that price and value are diverging from each other over the sample period but are still statistically cointegrated. I examine this issue further in my vector-based analysis in Section IV.

To summarize the results outlined in this section, I provide evidence consistent with a significant change in the long-run relation between price and accounting fundamentals when comparing the recent 1994–2008 period with the earlier 1979–1993 period. The results are stronger when the regression model includes both a constant and a trend term, which suggests that, in the earlier 1979–1993 period, the inclusion of a time-trend is important in finding evidence of co-movement between price and accounting fundamentals. In the latter period, however, price and

accounting fundamentals are not cointegrated in either of the specifications (with or without a trend), documenting that prices and accounting fundamentals do not co-move in recent years.

IV. FORECASTING AND MULTIVARIATE ANALYSIS

Univariate Forecasts of Index-Level Returns and Changes in Accounting Fundamentals

Lee et al. (1999) provide evidence that the cointegration between accounting fundamentals and price is associated with the correction of mispricing by documenting that the ratios of accounting fundamentals-to-price predict future returns. Thus, the lack of co-movement in recent periods documented in the prior section could have implications for the ability of accounting fundamental-to-price ratios to predict future returns.

Panel A of Table 3 reports summary statistics from a regression of the average index-level return over the subsequent K periods on the fundamental-to-price ratio, $vf(x)$, from the current period. I report only results for the $vf(x)$ ratio, as Lee et al. (1999) present evidence that the dividend- and book-value-based ratios have little predictive ability over future returns. Specifically, I estimate the following regression:

$$\sum_{k=1}^K Ret_{t+k}/K = a + b \left[\frac{vf(x)}{p} \right] + e_t, \quad (16)$$

where Ret_{t+k} is the cumulative return on the CRSP value-weighted index over the next K periods, and $K = 1, 3, 6, 9, 12$, and 18 months. For the models where $K > 1$, the power of the test is increased when using overlapping observations (Campbell 1993); as such the dependent variable overlaps by $K-1$ periods inducing serial correlation in the residuals. Accordingly, following Lee et al. (1999, 1716), I estimate these regressions using the procedure in Hansen and Hodrick (1980), which is a Generalized Method of Moments (GMM) estimate of the standard errors with a Newey-West correction (Newey and West 1987). As in Lee et al. (1999), if price is responding to the difference between the current and stable level of the fundamental-to-price ratio, then the slope coefficient, b , should be positive and significant.

In Table 3, I report the results from these regressions. For the 1979–2008 period, I do not find any evidence of predictability of future returns based on current $vf(x)/p$ levels, with all of the slope coefficients statistically insignificant in Column (2). When I examine the historical period 1979–1993 only, however, I find evidence of predictability of future returns using current $vf(x)/p$ levels, consistent with the evidence documented in Lee et al. (1999). Specifically, in Column (5), the slope coefficient, b , is decreasing over the holding period of the return (for example, for $K = 1$, $b = 0.072$; while for $K = 12$, $b = 0.053$) and the adjusted R^2 is increasing over time. The positive and statistically significant slope coefficients are evidence of the usefulness of $vf(x)/p$ in predicting aggregate market returns in the earlier time period. These results are consistent with a model of cointegration between price and accounting fundamentals, where revisions of market prices play a central role in keeping prices aligned with accounting fundamentals.

In stark contrast, in the recent period 1994–2008, I do not find evidence that the $vf(x)/p$ ratio is statistically positively associated with future aggregate returns at conventional levels. While the coefficient estimates in this period are negative, they are not statistically significant. In Column (10), I report Chow tests to formally test for structural change, all of which provide evidence of a structural change between the historical period (1979–1993) and the recent period (1994–2008) at $p < 0.001$. These results are consistent with a breakdown in the model of cointegration, where revisions of market prices are not related to the alignment of price levels with accounting fundamentals.

TABLE 3

Forecasting CRSP Value-Weighted Returns Using $v\hat{f}(x)/p$ and Tests for Structural Change

$$\sum_{k=1}^K \text{Ret}_{t+k}/K = \alpha + b[v\hat{f}(x)/p] + e_t. \quad (16)$$

Forecast										
1979–2008				1979–1993			1994–2008			Chow Test
<i>K</i>	(1) a	(2) b	(3) Adj. R ²	(4) a	(5) b	(6) Adj. R ²	(7) a	(8) b	(9) Adj. R ²	(10) F-statistic
1	0.011 ^δ (1.84)	0.003 (0.24)	−0.003	0.037** (3.64)	0.072* (2.41)	0.046	−0.003 (−0.40)	−0.018 (−1.25)	0.009	7.25**
3	0.010 ^δ (1.77)	0.002 (0.19)	−0.002	0.036** (4.48)	0.070* (2.99)	0.137	−0.003 (−0.42)	−0.018 (−1.43)	0.033	19.92**
6	0.012* (2.13)	0.006 (0.53)	0.003	0.034** (4.95)	0.064** (3.57)	0.226	0.001 (0.13)	−0.011 (−0.92)	0.022	27.71**
9	0.013* (2.25)	0.007 (0.64)	0.009	0.032** (5.15)	0.057** (3.70)	0.268	0.003 (0.43)	−0.007 (−0.56)	0.010	28.89**
12	0.014* (2.50)	0.008 (0.74)	0.015	0.030** (6.12)	0.053** (4.34)	0.312	0.005 (0.68)	−0.005 (−0.39)	0.003	28.87**
18	0.016* (3.26)	0.012 (1.09)	0.043	0.027** (6.62)	0.041** (4.76)	0.332	0.008 (0.94)	−0.001 (−0.05)	−0.006	20.95**

*, ** p < 0.05 and p < 0.001, respectively.

^δ p < 0.1.

This table summarizes the changes in the forecasting ability of the value-to-price ratio over future index level returns. I use CRSP value-weighted returns as the dependent variable, and I use the aggregate value model $v\hat{f}(x)/p$, which is the one-period forecast-based residual income model using historical earnings-to-price ratio, in logs. For the forecasting periods when $K > 1$, the dependent variable overlaps for $K-1$ periods, and as such the reported Z-statistics in parentheses are computed using the generalized methods of moments (GMM) estimator with the Newey-West correction for the $K-1$ overlapping periods and general heteroscedasticity. Future return data are available until December 2009; hence, due to data availability, the 18-month-ahead models include only $v\hat{f}(x)/p$ observations until June 2008 ($T = 354$ in columns 1–3, $T = 174$ in columns 6–9). The Chow test is a test for structural change and is estimated with a breakpoint at December 1993; the test statistic follows the F-distribution.

Vector-Based Analysis of the Value-to-Price Relation

Since cointegration, a test of co-movement between two or more variables, is inherently multivariate, further analysis using systems of equations is particularly useful in understanding the dynamic nature of the relation between prices and accounting fundamentals. While the estimation procedures in this section are more complex and require the estimation of additional parameters, the benefit of a multivariate approach is that it enables refinement of the interpretation of the results in the prior section. Specifically, it enables me to test the possibility that price is cointegrated, or co-moving, with accounting fundamentals but with a parameter that is significantly different from 1; this approach also allows for further investigation of the strong time-trends observed in the data.

In matrix form, the process of price and value can be written as a vector autoregression (VAR) in levels (matrices are represented in boldface):

$$x_t = v + A_1 x_{t-1} + \varepsilon_t, \quad (17)$$

where $x_t = (f_t, p_t)'$ is a 2×1 vector that includes price and an accounting fundamental variable, v is a 2×1 vector of constant terms, A_1 is a 2×2 matrix of parameters that relate current accounting fundamentals and prices to each of the lags of accounting fundamentals and of prices, and ε_t is a 2×1 vector of error terms that are mean-zero multivariate normal disturbances with variance Ω .

Under certain assumptions, the Granger representation theorem of [Engle and Granger \(1987\)](#) states that a cointegrated process can be represented as a vector-error correction model (VECM). In Appendix A, I show that, following the Granger representation theorem, Equation (18) can be written as a VECM including an intercept and time-trend as follows:

$$\Delta x_t = \alpha(\alpha_0 + \beta x_{t-1}) + \sum_{i=1}^{p-1} \Gamma_i \Delta x_{t-i} + \varepsilon_t, \quad (18)$$

$$\Delta x_t = \alpha(\alpha_0 + \delta t + \beta x_{t-1}) + \sum_{i=1}^{p-1} \Gamma_i \Delta x_{t-i} + v + \varepsilon_t, \quad (19)$$

where $a = \alpha\beta' = A_1 - I_k$ (I_k is the identity matrix), $\beta = (\alpha_0, \delta, \beta)'$, $\Gamma_i = \sum_{j=i+1}^{p-1} A_j$, A_j is a $2 \times p$ matrix of parameters that relate current changes in accounting fundamentals and prices to each of the p lags of changes in accounting fundamentals and changes in prices, and v and ε_t are defined as in Equation (17). Importantly, if the lagged levels term, $\alpha(\alpha_0 + \delta t + \beta x_{t-1})$, is omitted for cointegrated variables, a VAR model in changes is misspecified ([Engle and Granger 1987](#)). Following [Johansen \(1988\)](#), I test the model by examining the rank of the matrix a , which provides the following information: (1) if a has full rank ($r = k$), then all components of x_t are $I(0)$ stationary, or (2) if the rank of the matrix is $r < k$, then there are r stationary cointegrating relations. To assess the statistical significance of the rank of the matrix, I report two test statistics, the *Trace* and the *Max* statistics based on the eigenvalues, or characteristic roots, of the matrix a . In this case, the expected number of cointegrating relations equals 1, as I expect one linear combination between price and accounting fundamentals to remove the common trend of intrinsic value. To find evidence of cointegration, the *Trace* and *Max* statistics should reject the $rank = 0$ but not reject the $rank \leq 1$ ([Johansen and Juselius 1990](#)). Alternatively, when there is no evidence of cointegration, then the $rank = 0$ will not be rejected by either statistic.

Table 4 presents the estimation of a VECM model of Equation (18) that includes a constant in the long-run relation between price and fundamental value, and the estimation of the model that also includes a time-trend in the long-run relation represented as Equation (19). I report estimates of the model using aggregate price and aggregate $yf(x)$. As the indices are in levels, I CPI-adjust the aggregate price and accounting fundamentals to 2008 dollars and then log the indices.²⁰ Consistent with the results reported in Table 2, Table 4 documents statistical evidence of cointegration only in the earlier 1979–1993 period. For example, in the 1979–1993 period, for the model including a constant in the long-run relation (reported in Column (3) of Table 4), I find evidence of cointegration: both the *Trace* (25.020) and *Max* (21.543) statistics are significant at the 5 percent confidence level for tests of $rank = 0$, but are not significant at conventional levels for $rank = 1$.

The estimates of the cointegrated models of price and accounting fundamentals for the period

²⁰ To maintain consistency with the ratio based tests in the prior section, I estimate the model with two lags in the underlying VAR model. The underlying VAR has R^2 's greater than 98 percent for all models. The selection of the lag length using the minimum Schwarz Bayesian Information Criterion (BIC) and the Hannan-Quinn information criterion selected a single lag when I include up to 12 lags. The results are not qualitatively different when I use a single lag in the VAR model or when I use up to 12 lags. Results are quantitatively similar when the model is supplemented with analysts' forecasts.

TABLE 4
Johansen Tests for Cointegration between Fundamentals and Price

Constant:

$$\Delta x_t = \alpha(\alpha_0 + \beta x_{t-1}) + \sum_{i=1}^{\rho-1} \Gamma_i \Delta x_{t-i} + \varepsilon_t. \quad (18)$$

Trend:

$$\Delta x_t = \alpha(\alpha_0 + \delta t + \beta x_{t-1}) + \sum_{i=1}^{\rho-1} \Gamma_i \Delta x_{t-i} + v + \varepsilon_t. \quad (19)$$

	1979–2008 <i>T</i> = 360		1979–1993 <i>T</i> = 180		1994–2008 <i>T</i> = 180	
	(1) Constant	(2) Trend	(3) Constant	(4) Trend	(5) Constant	(6) Trend
Long-Run Parameters						
α_0	–1.208	–18.268	–4.743**	–6.570	–18.462 ^δ	–66.065
β	–0.954*	0.277	–0.663**	–0.536*	0.123	3.616
δ		–0.009*		–0.001		–0.032**
Error-Correction Parameters (α)						
α_v	–0.007*	–0.043*	–0.036	–0.075*	–0.011*	–0.091*
α_p	–0.004	–0.011	0.215**	0.207*	–0.010*	0.038
Short-Run Parameters (Γ)						
v_v		–0.001		0.005*		0.002
$\gamma_{v,p}$	–0.128*	–0.112*	–0.039	–0.050	–0.220*	–0.178 ^δ
$\gamma_{v,v}$	0.192**	0.199**	0.177*	0.186*	0.207*	0.233*
v_p		0.002		0.002		0.005
$\gamma_{p,v}$	0.045	0.044	–0.032	–0.024	0.014	0.001
$\gamma_{p,p}$	0.092 ^δ	0.098 ^δ	0.106	0.084	0.101	0.118
Tests of Cointegration						
<i>Trace rank</i> = 0	14.401	15.089	25.020*	32.350*	13.411	17.415
<i>Trace rank</i> = 1	3.197	2.911	3.477	10.762	3.449	6.322
<i>Max rank</i> = 0	11.204	12.178	21.543*	21.588*	9.971	11.094
<i>Max rank</i> = 1	3.197	2.911	3.477	10.762	3.449	6.322

*, ** $p < 0.05$ and $p < 0.001$, respectively.

^δ $p < 0.1$.

This table summarizes the results of vector-error correction models (VECM) and reduced rank statistics that provide a test of cointegration (Johansen 1988). I provide estimates of two alternative forms of the VECM between price and value. In columns under the heading “Constant,” I report estimates of Equation (18); in columns under the heading “Trend,” I report estimates of Equation (19). The two series used in the model are the times-series of aggregate market value (p) and the aggregate accounting-based measure of value ($v_f(x)$), where value is measured using the sum of book-value and a perpetuity of one-period-ahead residual income value, based on historical earnings, discounted using the three-year Treasury constant-yield-to-maturity bond rate plus a 6 percent equity premium. Each index is constructed as an aggregate index reported in millions of U.S. dollars; the index is then inflation-adjusted to 2008 prices and logged. The reduced rank statistics reported are the trace test (*Trace*) and the maximum eigenvalue test (*Max*). Cointegration is confirmed in these tests when the test of the null *rank* = 0 is rejected, but the test of the null *rank* = 1 is not rejected.

1979–1993, with a constant in Column (3), and with a constant and trend in Column (4), provide insights into the stable relation between price and accounting fundamentals that can be compared to the parameters estimated from more recent periods. The first is that the estimate of the long-run relation between price and accounting fundamentals is significantly different from -1 , as assumed in the tests for stationarity of ratios of fundamentals-to-price. For example, in Column (3) the estimate of β_1 is -0.663 , and in untabulated results, the 95 percent confidence interval for β_1 is -0.744 to -0.582 , which provides statistical evidence to reject a long-run relation between price and $vf(x)$ of -1 at the 5 percent level.²¹ In Column (4), I report the model that includes both a constant and a trend. In this model, the β_1 estimate of -0.536 is again statistically different from 1 with a 95 percent confidence interval of -0.765 to -0.308 (untabulated).

One can obtain further information from the model by examining the error-correction parameters. Specifically, within a set of cointegrated variables, at least one will respond to the disequilibrium in the system. A summary of the responsiveness of the variables is provided by the model's error-correction parameters. The response parameter for price in Column (3), α_p , is 0.215 and statistically significant at $p < 0.001$. The response parameter for $vf(x)$, α_v , is not statistically significant at conventional levels. Consistent with the forecasting regressions in Table 3, this result suggests that, when cointegrated, price responds to the disparity from the equilibrium relation between price and accounting fundamentals. The error-correction parameters for the model that includes a trend, reported in Column (4), support a similar interpretation, except that there is also weak evidence that accounting fundamentals respond to the disparity. In general, the strong error-correction parameter for price supports Penman's (2003, 77) assertion that one of accounting's roles is to "challenge speculative beliefs, and so anchor investors on fundamentals."

In direct contrast, the parameters for the more recent 1994–2008 period do not reject the null of no cointegrating relations ($r = 0$) according to either the *Trace* or *Max* statistics; see Columns (5) and (6). These parameters require care in interpretation, however, as the model is not stable. The most striking difference between the models reported in Columns (5) and (6) is that the long-run relation β is not statistically different from zero, which implies no relation between price and accounting fundamentals in recent periods. This lack of association is explained by the trend term, δ , in Column (6) of -0.032 , which suggests an increasing disparity between price and accounting fundamentals over time. Another notable difference appears in Column (5), where the error-correction parameter of -0.010 for price, α_p , is negative and statistically significant at $p < 0.05$. This suggests that changes in price are moving away from the long-run relation (which in this case is a constant difference), consistent with the unstable relation between p and $vf(x)$. When a trend is included in Column (6), α_p is positive but not significant at conventional levels.

Overall, estimates of the long-run parameters provide evidence of a stable long-run relation between price and accounting fundamentals from 1979–1993, but no significant long-run relation is evident in the recent 1994–2008 period. In the stable, cointegrated time period, I find evidence consistent with the notion that fundamentals provide an anchor to prices, based on the error-correction parameter for price. In the recent period, however, there is no evidence of anchoring. The estimation of the additional parameters in this section also highlights the importance of recognizing the role played by time-trends in recent time periods. The implications of strong independent time-trends include the possibility of introducing a spurious bias in the parameters estimated using more traditional OLS-based analyses (see also Callen and Morel 2005).

²¹ More technically, the cointegrating vector $a = [1, -1]'$ is not supported by the data.

V. ROBUSTNESS ANALYSIS

Robustness to Supplementing Accounting Fundamentals with Analysts' Forecasts

In this subsection I present tests to investigate the possibility that measurement error in accounting fundamentals has increased over time by supplementing accounting fundamentals with analysts' forecasts. As analysts can incorporate information in press releases into their forecasts, the use of analysts' forecasts likely provides a better estimate of expected earnings, which is increasingly important given the increasing frequency with which information is disclosed by firms in press releases and conference calls (Brown and Rozeff 1978; Dechow et al. 1999).

I present stationarity tests for fundamental-to-price ratios that include analysts' forecasts rather than historical earnings. In this analysis, I restrict my sample to firms covered by I/B/E/S. I supplement the main model $vf(x)$, substituting analysts' forecasts with historical earnings in Equation (11). I label this model $vf(f)$.²² I supplement the linear dynamics model, $vl(x)$, with analysts' forecasts following Dechow et al. (1999, 7), who assume that the "other information" term can be represented as:

$$v_t = f(1)_t - r \cdot b_t - \omega_t x_t^a,$$

which allows me to estimate the model as:

$$vl(f)_t = \frac{b_t + \omega_t x_t^a}{(1 + r_t - \omega_t)} + \frac{(1 + r_t)v_t}{(1 + r_t - \omega_t)(1 + r_t - \gamma_t)}, \quad (20)$$

where the terms are defined in Equations (12) and (15).

To the extent that the consensus I/B/E/S forecasts of earnings do not rely on conservative treatments of accounting, then it possible that an EPS-based model, such as that proposed by Ohlson and Juettner-Nauroth (2005), may provide a better measure of value. The capitalization of forward earnings is also related to the linear information dynamics of Ohlson (1995) and results from assuming that residual income has zero persistence and that other information is a random walk. I estimate the earnings capitalization model (ve) as $ve(1)_t = \frac{f(1)_t}{r_t}$. I also provide an estimate of the Ohlson-Juettner model following Gode and Mohanram (2003):

$$ve(f)_t = \frac{f(1)_t}{r_t} + \frac{stg_t - r_t(f(1)_t - d(1)_t)}{(r_t - ltg_t)r_t}, \quad (21)$$

where the model expresses value as a function of forecast earnings per share, $feps$; the required rate of return, r ; short-term growth in earnings, stg ; and long-term growth, ltg . I model the short-term growth rate, stg , using the average of $(feps_2 - feps_1)/feps_1$ and the long-term growth estimate reported by I/B/E/S. Observations without I/B/E/S growth estimates are replaced with zero growth. All other terms are as defined above.

In both Panel A and Panel B of Table 5, my results are generally consistent with the main analysis. Specifically, all of the ratios are nonstationary in the 1994–2008 period. Except for the linear dynamics model, $vl(f)$, all of the ratios are stationary in the 1979–1993 period and are structurally different across the two time periods according to the Chow test. All models that include a trend are consistent with the main results. There is some evidence of stationarity for the $ve(f)$ -to-price ratio over the full 1979–2008 period; however, in the recent period, there is no evidence of stationarity for this ratio, and there is evidence of a structural change in the times-series

²² Changing the forecast horizon by adding additional forecasts to the model, as in Lee et al. (1999) and Frankel and Lee (1998), requires assumptions regarding payout policy, and does not change the main results.

TABLE 5

Analysis of Supplementing Accounting Fundamentals with Analysts' Forecasts

Panel A: Phillips-Perron Tests for Stationarity with a Constant

$$\Delta \varepsilon_t = \alpha_0 + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (7)$$

Variable	1979–2008 <i>T</i> = 360		1979–1993 <i>T</i> = 180		1994–2008 <i>T</i> = 180	
	(1)	(2)	(3)	(4)	(5)	(6)
	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)
$vf(f)/p$	−3.47	−0.71	−24.57**	−3.58**	1.08 [†]	0.38
$vl(f)/p$	−4.39	−1.48	−5.34	−1.54	0.81	0.27
$ve(1)/p$	−9.46	−1.71	−33.38**	−4.29**	0.30 [†]	0.10
$ve(f)/p$	−18.88*	−2.97*	−31.13**	−4.64**	1.61 [†]	0.59

Panel B: Phillips-Perron Tests for Stationarity with a Constant and a Time-Trend

$$\Delta \varepsilon_t = \alpha_0 + \delta t + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (8)$$

Variable	1979–2008 <i>T</i> = 360		1979–1993 <i>T</i> = 180		1994–2008 <i>T</i> = 180	
	(1)	(2)	(3)	(4)	(5)	(6)
	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)
$vf(f)/p$	−3.56	−0.74	−26.00*	−3.58*	−3.23 [†]	−1.00
$vl(f)/p$	−1.16	−0.27	−25.38*	−3.65*	−1.79 [†]	−0.58
$ve(1)/p$	−8.59	−1.56	−36.37**	−4.37**	−3.80 [†]	−1.19
$ve(f)/p$	−25.04*	−3.46*	−52.03**	−5.82**	−1.64 [†]	−0.56

*, ** $p < 0.05$ and $p < 0.001$, respectively.

[†] $p < 0.1$.

[†] Indicates that the parameter estimates are significantly different (at the $p < 0.05$ level) in the 1994–2008 period relative to the 1979–1993 period according to the Chow test.

$vf(f)$ is the one-period forecast-based residual income model using analysts' forecasts when available, $vl(f)$ is the linear information dynamics model with analyst forecasts as "other information" from [Dechow et al. \(1999\)](#), $ve(1)$ is the earnings capitalization, and $ve(f)$ is the [Ohlson and Juettner-Nauroth \(2005\)](#) model of earnings growth. In Panel B, the fundamental-to-price ratios used in this analysis are the $vf(x)/p$ and $vl(f)/p$ models. Firms are assigned to growth quintile portfolios annually in January. The test is of the null of a unit root in the times-series ($\rho = 1$). The two test statistics from the Phillips-Perron regressions are an adjusted regression coefficient Z-Rho (Z_ρ) and an adjusted t-statistic Z-Tau (Z_τ).

according to the Chow test. These results suggest that supplementing accounting fundamentals with analysts' forecast information is not sufficient to explain the lack of co-movement between prices and accounting fundamentals observed in recent periods.

Robustness to Portfolio Formation on Risk and Growth Variables

If the lack of cointegration in recent periods is solely due to the misspecification of firm-specific rates of return or expected growth, then aggregating firms into portfolios of similar risk levels, or expected growth levels, should result in stationary fundamental-to-price ratios. In this subsection, I provide a robustness analysis of the stationarity of the log fundamental-to-price ratio to the aggregation of firms into portfolios based on risk and growth characteristics. In Panel A of Table 6, I

report a risk-based test for potential measurement error in the expected rate of return by forming quintile portfolios of $v_f(x)/p$ based on the level of the firm's beta. The portfolio approach is preferable to adjusting the firm-specific cost of capital due to the measurement errors in firm-specific estimates of beta, which are mitigated by using ranks. I estimate beta over the prior 12 months and assign firms annually to their quintile portfolios at the beginning of January for each year. When examining the recent period, 1994–2008, I do not find support for this risk-based explanation, as none of the portfolios reject the null of nonstationarity. In the historical period, evidence of stationarity is reliably observed for the higher beta quintiles, and weakly observed for the lower beta quintile portfolios.²³

In Panel B of Table 6, I present tests to investigate the role of potential measurement error in accounting fundamentals due to differences in expected growth by sorting firms into portfolios based on long-term analysts' earnings per share forecasts. Again, the portfolio approach is preferable to adjusting the valuation model due to the measurement errors in firm-specific growth estimates and the requirement that the growth rate be lower than the cost of capital. In Panel B, consistent with the main results, there is no reliable evidence of cointegration between price and accounting fundamentals for any of the portfolios during the period 1994–2008. Consistent with the main results, in the 1979–1993 period, Quintiles 2 through 5 are stationary. Quintile 1, which is nonstationary, however, includes firms with no I/B/E/S long-term growth estimates, suggesting that the fundamentals hypothesis may not generalize to smaller firms. While it is possible that price diverged from fundamentals due to various combinations of discount rates and growth expectations, the results reported in this subsection do not support either of these explanations.

Further Analysis of the Timing of the Breakdown in Cointegration

In this subsection, I provide additional robustness tests to investigate the stability of the ratio at different breakpoints in the times-series. In the main analysis, I focus on splitting the sample period into two equal time periods, 1979–1993 and 1994–2008, to control for power issues relating to the number of observations in each period. To assess the robustness of these results, and to examine whether a more timely test for identifying the breakdown in cointegration is possible, I perform an iterative analysis by adding a single observation to the times-series beginning in April 1987 (allowing a minimum of 100 observations for these tests, and increasing by one observation per iteration). Cointegration between price and accounting fundamentals is first observed weakly in September 1990, and March 1996 is the first time cointegration is not statistically present at the $p < 0.10$ level. After October 1996, statistical evidence of cointegration is not reliably observed for tests on times-series that iteratively add a single month (without removing any observations) to the times-series for the remainder of the sample period.²⁴

²³ In untabulated analyses, I relax the assumption that the equity premium equals 6 percent. Equity premia ranging from 0 percent to 12 percent are generally consistent with my main analysis, with evidence of cointegration typically found only in the historical 1979–1993 period. Z-Rho is lowest with the equity premium set at 4 percent, which is close to the estimates of the equity premium in [Claus and Thomas \(2001\)](#). In the recent period, there is no evidence of a reduction in Z-Rho, when using lower equity premia suggesting that the assumption of lowering the discount rate does not improve the tendency of prices and accounting fundamentals to co-move. Similar to [Lee et al. \(1999\)](#), when I use constant discount rates equal to 6 and 12 percent, I find no evidence of cointegration in any of the three time periods examined. I obtain similar results when supplementing the model with analysts' forecasts.

²⁴ Results are similar when I include analysts' forecasts in the model; however, in this specification, the Z-Rho test statistic is under 10 percent for the periods starting in 1979 and ending in April through November 2007, but the Z-Tau statistic remains above 10 percent. This provides some very weak evidence that there was a period when price and accounting fundamentals supplemented with analysts' forecasts co-moved in a similar fashion to the historical relation in 2005–2007. This conjecture cannot be confirmed, however, possibly as the time period is too short to provide a powerful test for cointegration. Similar results are found for rolling windows of 100 observations.

TABLE 6

Further Analysis of Expected Rates of Return and Growth

Panel A: Phillips-Perron Tests for Firms Sorted into Portfolios by Their Market Beta

$$\Delta \varepsilon_t = \alpha_0 + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (7)$$

<i>vf(x)/p</i>	(1) 1979–2008 <i>T</i> = 360		(2) 1979–1993 <i>T</i> = 180		(3) 1994–2008 <i>T</i> = 180	
	(1)	(2)	(3)	(4)	(5)	(6)
	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)
Q1—Low <i>Beta</i>	−8.96	−1.55	−12.02 ^δ	−2.57	0.70	0.17
Q2	−4.02	−0.77	−11.96 ^δ	−2.47	0.93	0.26
Q3	−4.61	−0.90	−12.55 ^δ	−2.58 ^δ	1.62	0.49
Q4	−12.19 ^δ	−1.94	−23.04**	−3.45*	−2.45 [†]	−0.58
Q5—High <i>Beta</i>	−15.29*	−2.17	−20.54**	−3.27*	−4.22	−0.85

Panel B: Phillips-Perron Tests for Firms Sorted into Portfolios by Analysts' Long-Term Growth Estimates

$$\Delta \varepsilon_t = \alpha_0 + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (7)$$

<i>vf(f)/p</i>	(1) 1979–2008 <i>T</i> = 360		(2) 1979–1993 <i>T</i> = 180		(3) 1994–2008 <i>T</i> = 180	
	(1)	(2)	(3)	(4)	(5)	(6)
	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)	Z-Rho (Z_ρ)	Z-Tau (Z_τ)
Q1—No Growth	−3.56	−0.66	−10.57	−2.31	2.22 [†]	0.61
Q2—Low Growth	−2.70	−0.53	−14.50*	−2.98*	2.29 [†]	0.73
Q3	−1.67	−0.36	−31.32**	−4.03**	0.76 [†]	0.28
Q4	−0.46	−0.1	−35.38**	−4.24**	2.55 [†]	0.95
Q5—High Growth	−7.72	−1.53	−13.76*	−2.61 ^δ	−1.37 [†]	−0.42

*, ** $p < 0.05$ and $p < 0.001$, respectively.

^δ $p < 0.1$.

[†] Indicates that the parameter estimates are significantly different (at the $p < 0.05$ level) in the 1994–2008 period relative to the 1979–1993 period according to the Chow test.

This table summarizes the robustness of results in Table 2 to different discount rates and market risk-based portfolios. The unit root tests are performed without a time-trend, testing Equation (6). The test variable in these models is $vf(x)/p$. In Panel A, *Beta* is estimated using the association between the firm's return and the value-weighted index on CRSP, both less the risk-free (T-bill) rate and estimated over the prior 12 months using monthly data. In Panel B I use analysts' forecasts of long-term growth to form quintile portfolios. Firms are assigned to quintile portfolios annually in January. In both panels, the test is of the null of a unit root in the times-series ($\rho = 1$). The test statistics from the Phillips-Perron regressions are an adjusted regression coefficient Z-Rho (Z_ρ) and an adjusted t-statistic Z-Tau (Z_τ).

These results provide some evidence that the methods used in this study could potentially be used to help identify periods of increasing speculative movements in price relatively early in the divergence of price from accounting fundamentals. The breakdown in cointegration during 1996 could have been seen as an early warning sign, rather than a sign that the economy was entering a “new paradigm,” as many believed at the time. For example, Robert Rubin, then U.S. Treasury Secretary, stated in October 1997 that the fundamentals of the American economy were sound

(Chancellor 2000, 229). While my interpretation of this evidence is with hindsight, these tests do provide the first step toward predicting periods of larger speculative movements in price.

VI. CONCLUSION

Historically, the relation between price and accounting fundamentals has been one of co-movement, implying that price anchors to the fundamentals of the firm as represented in accounting reports and that any mispricing of this information is short-lived. Statistical evidence of co-movement, or cointegration, of prices with accounting fundamentals is important, as it rules out extended speculative price movements, such as those labeled “bubbles,” and it is consistent with prices anchoring to accounting fundamentals (Penman 2003). I investigate whether a cointegrating relation is present in recent data, as there is an ongoing debate regarding the existence of excessive speculation during recent periods. Following Harrison and Kreps’s (1978) definition of speculation as the component of price that does not co-move with fundamental value, I test whether models of fundamental value that use accounting fundamentals are cointegrated with market prices in recent periods.

I find that evidence of cointegration between price and accounting fundamentals is not reliably observed for the period 1979–2008. Cointegration is observed only in the earlier half of the times-series, from 1979–1993. In the subsequent half of the times-series, from 1994–2008, I do not find any reliable evidence of cointegration between price and accounting fundamentals. The results are robust to alternative models used to combine accounting fundamentals, including supplementing accounting fundamentals with analysts’ forecasts. Like Lee et al. (1999), I find that, historically, time-variation in interest rates has been crucial in finding evidence of co-movement between accounting fundamentals and prices. In recent periods, however, the data do not support risk and growth explanations for the divergence of prices from accounting fundamentals.

I also provide additional analysis of the forecasting properties of the fundamental-to-price ratio, along with multivariate analysis of the cointegrating relation, suggesting that the historically observed short-term tendency of prices to correct for the disparity between price and accounting fundamentals is not observed in recent periods. Taken together, the results of this study suggest that the significant change in the time-series dynamics of price is the driver of the lack of cointegration between price and accounting fundamentals.

These results provide an important first step in understanding the role of accounting fundamentals in recent time periods. Specifically, my results contribute to the policy debate on whether large shifts in prices are due to shifts in fundamentals or due to departures of prices from fundamentals; the results appear to be more consistent with the latter explanation. My results also have implications for researchers examining the properties of aggregate accounting fundamentals. Specifically, the formal statistical tests for co-movement that I utilize suggest that in recent periods, while the upward time-trends in both prices and accounting fundamentals appear visually similar, statistically they are not linearly related, implying that linear estimates of the association between accounting fundamentals and market prices are likely to be spurious.

An important caveat applies to these findings. It is possible that, in recent periods, investors have conditioned their expectations on measures of intrinsic value not captured by my measures of fundamental value (Hamilton and Whiteman 1985). My results do not rule out this possibility; rather, I show that the combination of accounting fundamentals, analysts’ forecasts, and other information considered in this study—estimation techniques that have been widely used in prior research—do not appear to be good proxies for the measures of intrinsic value used by investors to condition their trades over the last 15 years. One of the strengths of the residual-income-based models, however, is that they do have historical cointegrating relations with price, and they have a

strong theoretical tie to the market fundamentals hypothesis (i.e., present value of future cash flows). I leave to future research the question of whether the long-run association between market prices and accounting fundamentals is best explained by the traditional fundamentals hypothesis or by the many alternative hypotheses presented in the literature loosely labeled “behavioral finance.”

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APPENDIX A

FURTHER DESCRIPTION OF THE UNIVARIATE COINTEGRATING RELATION

This appendix presents some intermediate results leading to the tests of stationarity of the log fundamental value-to-price ratio. Rearranging Equation (4) yields $\varepsilon_t = f_t - \beta p_t$. For the restriction $\beta = 1$, Equation (4) has the univariate representation of the log fundamental value-to-price ratio as $f_t - p_t = (f_t/p_t) \equiv \varepsilon_t$ (Lee et al. 1999). In the first case, $\varepsilon_t = \alpha_0 + \varepsilon_{t-1} + e_t$, and in the latter, $\varepsilon_t = \gamma_0 + \gamma_1 t + e_t$. Following Bhargava (1986), one can write a general time-series process for ε_t combining these two possibilities as follows:

$$\varepsilon_t = \psi_0 + \psi_1 t + \rho[\varepsilon_{t-1} - \psi_0 - \psi_1(t-1)] + e_t. \quad (\text{A1})$$

Rearranging Equation (A1) yields:

$$\varepsilon_t = \alpha_0 + \delta t + \rho\varepsilon_{t-1} + e_t, \quad (\text{A2})$$

which is Equation (6) in the text, where $\alpha_0 \equiv \psi_0(1 - \rho) + \psi_1\rho$ and $\delta = \psi_1(1 - \rho)$. As the OLS estimate of Equation (A2) will be affected by serially correlated error terms, tests for a unit root are estimated in differences:

$$\Delta\varepsilon_t = \alpha_0 + \delta t + (\rho - 1)\varepsilon_{t-1} + e_t. \quad (\text{A3})$$

Restricting the time-trend $\psi_1 = 0$ in Equation (A3) yields Equation (7) in the text, with the unrestricted time-trend yielding Equation (8). In Equation (A3) the null hypothesis, $\rho = 1$, suggests that ε_t does not mean-revert, but wanders arbitrarily around a trend (i.e., $\Delta\varepsilon_t = \psi_1 + e_t$). The alternative hypothesis, $|\rho| < 1$, suggests that ε_t mean-reverts around the level $\alpha_0/(1 - \rho)$ or around a constant and time-trend, i.e., $(\psi_0 + \psi_1\rho)/(1 - \rho)$. As noted by Lee et al. (1999), rejection of the null hypothesis implies that fundamental value and price are cointegrated.

FURTHER DESCRIPTION OF THE MULTIVARIATE COINTEGRATING RELATION

Here I briefly summarize the relation between cointegration, common trends, and error correction models as intermediate results leading to the multivariate models presented as Equations (18) and

(19) in the text. If there is a $(g \times r)$ matrix a with reduced rank r , then the vector autoregressive series x_t is cointegrated with the cointegrating vector a and can be written as:

$$z_t = a'x_t, \quad (\text{A4})$$

where z_t is the matrix of equilibrium errors, which is stationary. These results can easily be generalized to include more than two variables in the vector x_t as long as the dimension of the x_t vector is the same as the dimension of the ε_t vector; i.e., they are both of dimension $K \times 1$, where K is the number of variables included in the x_t vector. An alternative case not examined here, where the dimension of the ε_t vector is less than the x_t vector, does not necessarily follow the model below (see [Ogaki 1998](#)).

The Granger Representation Theorem presented by [Engle and Granger \(1987\)](#) states that any cointegrated vector autoregressive process can be decomposed into a random walk, a stationary process, a deterministic component, and a term that depends on the initial values. The theorem also asserts the existence of a vector error correction representation for any cointegrated vector autoregressive process. I apply the theorem here to formalize the properties of Equations (18) and (19). By definition, any $I(1)$ process, when differenced, is stationary, $I(0)$. In differences, Equation (17) can be given the moving average representation:

$$\Delta x_t = \mu + \psi(L)e_t, \quad (\text{A5})$$

where μ is a matrix of deterministic trends such as a constant and time-trend; $\psi(L)$ is a matrix polynomial in the lag operator, with the lag at zero being the identity matrix, i.e.; $\psi(L) = I_k + \psi_1 L^1 + \psi_2 L^2 + \dots$, and the $(k \times 1)$ vector e_t is the linear forecast errors in x_t , given the lagged values of x_t . For convenience, let $\mu = 0$; then adding and subtracting $\psi(1)e_t$ to Equation (A5) yields $\Delta x_t = \psi(1)e_t + [\psi(L) - \psi(1)]e_t$; then integrating:

$$x_t = \psi(1)\xi_t + \psi^*(L)e_t + x_0, \quad (\text{A6})$$

where $\xi_t = \sum_{s=1}^t e_s$, yields $\psi^*(L) = (1 - L)^{-1} [\psi(L) - \psi(1)]$. This decomposition is often called the Beveridge-Nelson-Stock-Watson decomposition, in which the time-series process is composed of a permanent component, a random walk plus initial values, $\psi(1)\xi_t + x_0$, and a transitory component, $\psi^*(L)e_t$.

Combining Equations (A4) and (A6) yields a stationary representation of z_t as $z_t = \alpha'\psi(1)\xi_t + \alpha'\psi^*(L)e_t + \alpha'x_0$ as long as $\alpha'\psi(1) = 0$. This result can be interpreted as cointegration between two variables removing a common nonstationary, or non-mean reverting, component—in this case ξ_t , which for the models here is interpreted as intrinsic value. This is the key concept of cointegration: the linear combination of nonstationary processes removes a common nonstationary component, allowing for a stationary, or mean reverting, residual process. See [Stock and Watson \(1988\)](#) for further discussion of common trends in cointegrated processes.²⁵

Given these conditions, the autoregressive representation for Δx_t is, $\psi^*(L) \Delta x_t = -\alpha\beta'x_t + \varepsilon_t$, which is in VECM form ([Engle and Granger 1987](#)). Following [Johansen \(1988\)](#), this model allows for deterministic trends in both the mean of the cointegrating relation and the mean of the differenced series. The model is written as:

²⁵ In this paper, the common nonstationary component is intrinsic value. Systems such as these could be generalized to pairs or groups of time-series variables that are expected to co-move. Many accounting variables are nonstationary due to growth over time, as are prices in levels. Some other examples of possible processes include accruals and cash flows (through the common trend of earnings), and revenues and expenses (through matching). Finding cointegrated relations between various accounting variables may aid in forecasting pairs or groups of variables.

$$\Delta x_t = \alpha(\beta x_{t-1} + \mu + \rho t) + \sum_{i=1}^{\rho-1} \Gamma_i \Delta x_{t-1} + v + \tau t + \varepsilon_t, \quad (\text{A7})$$

where $\alpha\beta' = a$, $\mu + \rho t$ is a constant and time trend in the levels, and $v + \tau t$ is a constant and time-trend in the differences. As there is no basis for a quadratic time trend in the levels of the data, I restrict $\tau = 0$ and further restrict Equation (A7) by setting $\rho = 0$ and $v = 0$, which yields Equation (18) in the text. This specification allows the cointegrating relation to be stationary around a constant mean (μ). Removing restrictions on ρ and v yields Equation (19) in the text. In this case, the rejection of the null hypothesis of no cointegration suggests that the cointegrating equations are trend stationary.

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